DSP · L05

Frequency response

Cheat sheet · Day 07 · Fri 30 Oct · pack D4

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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FREQUENCY RESPONSE (L05 slides 27-35)
H(w) = sum_n h(n) e^{-jwn}        H(0)=sum h(n)    H(pi)=sum (-1)^n h(n)
LCCDE (all y on left): H(w) = sum_k b_k e^{-jkw} / (1 + sum_k a_k e^{-jkw})
  stable system only (poles inside |z|=1)
e^{jwn}  -> H(w) e^{jwn}
A cos(wn+phi) -> A|H(w)| cos(wn + phi + theta(w))     (sin stays sin)
Sum of sinusoids: do each frequency separately, add. DC: w=0. (-1)^n: w=pi.
Y(w)=H(w)X(w): LTI never creates new frequencies.
  New freq appears -> NOT LTI (nonlinear and/or time-varying).
  No new freq -> proves nothing.
One sinusoid pair gives H at that ONE frequency only.  delta(n) in -> h(n) out.
Symmetric FIR: factor out middle delay: 1+e^{-j2Lw} = 2e^{-jLw}cos(Lw)
  |H| = |real part|; negative real part -> add pi to phase.
1/(1-a e^{-jw}): |H| = 1/sqrt(1-2a cos w + a^2)
GEOMETRY: H = b0 e^{jw(N-M)} prod(e^{jw}-z_k)/prod(e^{jw}-p_k)
  |H| = |b0| prod V_k / prod U_k   (lengths: zero->e^{jw}, pole->e^{jw})
  theta = angle b0 + w(N-M) + sum angle(e^{jw}-z_k) - sum angle(e^{jw}-p_k)
  zero ON circle -> |H|=0 there; pole near circle -> sharp peak
  origin poles/zeros: no effect on |H|, only linear phase
  zero at z=1 kills DC; zero at z=-1 kills w=pi
SKETCH: walk e^{jw} from z=1 (w=0) to z=-1 (w=pi); check 0, pi, pole/zero angles.
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